Paths, combinations and Pascal

High school · counting · Progressive exercises · 05

Exercise 1. Find $\binom{8}{2}$ and explain why it equals $\binom{8}{6}$.

Exercise 2. Count monotone paths from $(0,0)$ to $(4,3)$.

Exercise 3. How many of these paths pass through $(2,1)$?

Exercise 4. Choose a team of three from eight people. Two specified people cannot be together. How many teams remain possible?